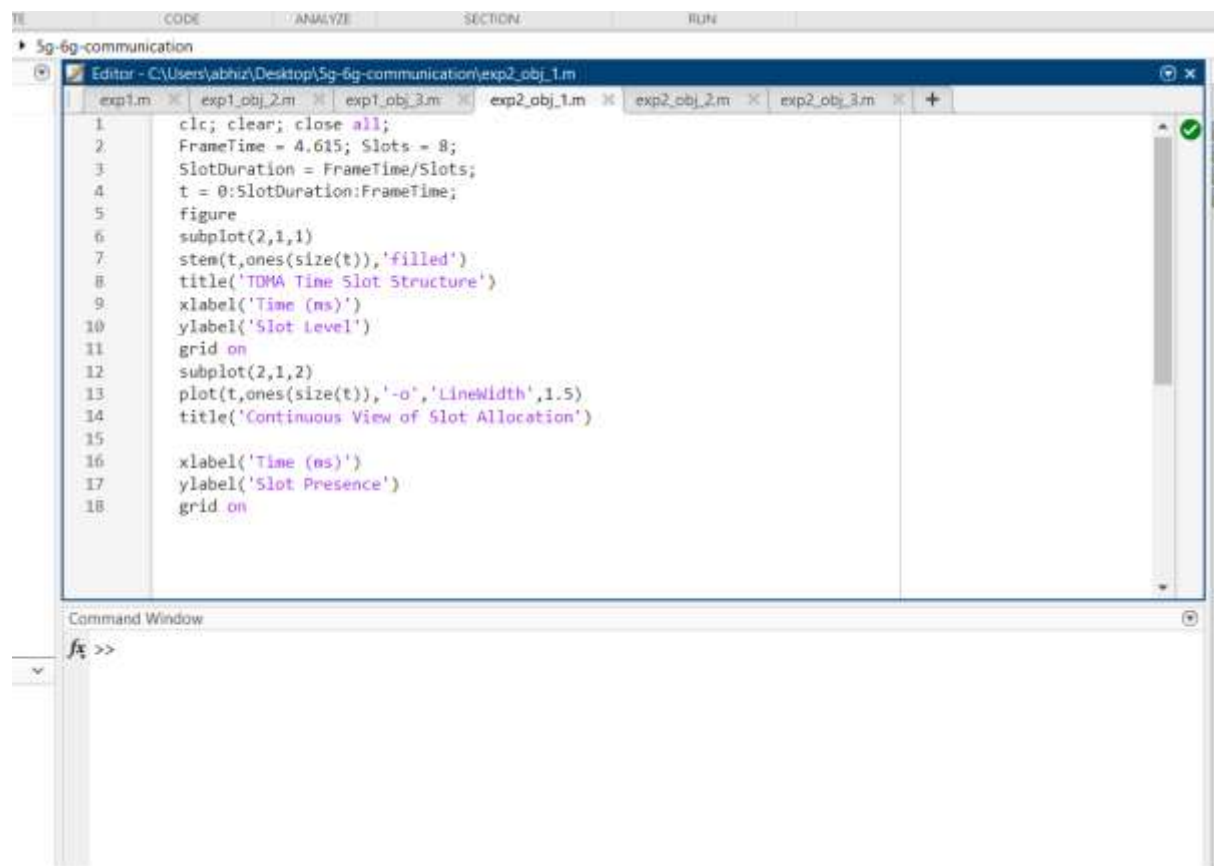


Exp2

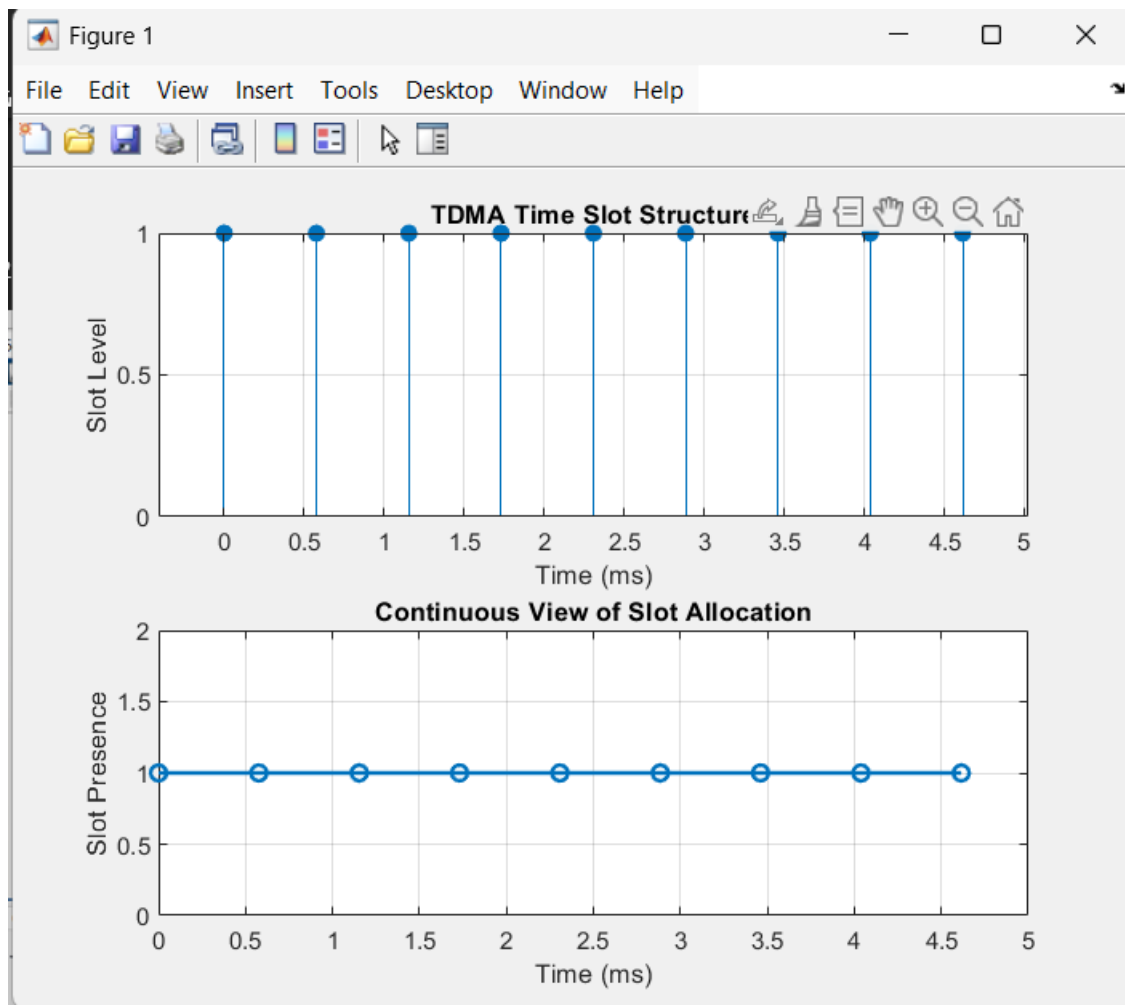


The image shows a MATLAB editor window titled "5g-6g-communication" with a file explorer on the left and a command window at the bottom. The editor displays a script named "exp2_obj_1.m" with the following code:

```
1 clc; clear; close all;
2 FrameTime = 4.615; Slots = 8;
3 SlotDuration = FrameTime/Slots;
4 t = 0:SlotDuration:FrameTime;
5 figure
6 subplot(2,1,1)
7 stem(t,ones(size(t)),'filled')
8 title('TDMA Time Slot Structure')
9 xlabel('Time (ns)')
10 ylabel('Slot Level')
11 grid on
12 subplot(2,1,2)
13 plot(t,ones(size(t)),'-o','LineWidth',1.5)
14 title('Continuous View of Slot Allocation')
15
16 xlabel('Time (ns)')
17 ylabel('Slot Presence')
18 grid on
```

The command window at the bottom shows the MATLAB prompt "f>>" and is currently empty.

Output



Exp2.2

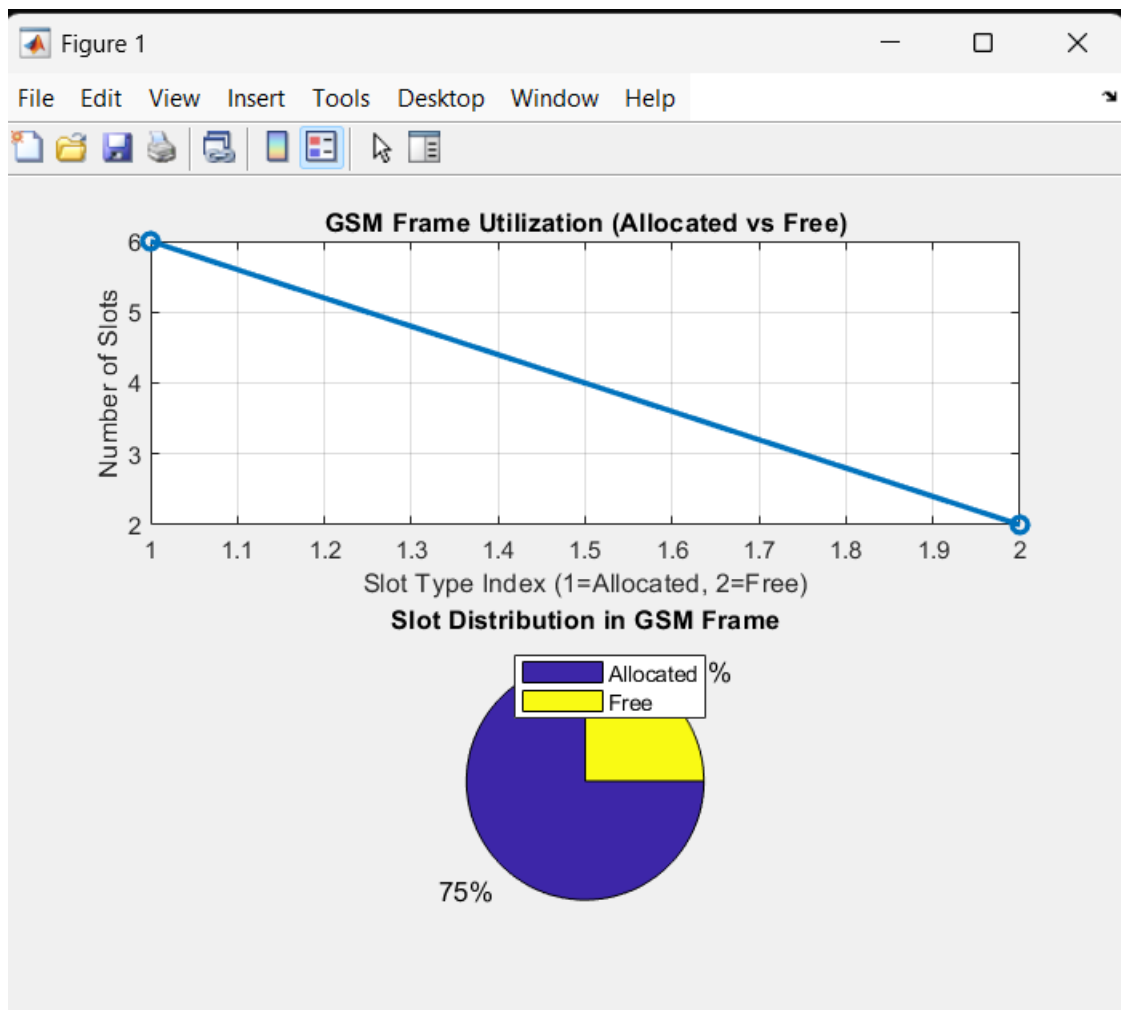
```
Editor - C:\Users\abhiz\Desktop\5g-6g-communication\exp2_obj_2.m
exp1.m x exp1_obj_2.m x exp1_obj_3.m x exp2_obj_1.m x exp2_obj_2.m x exp

1  clc; clear; close all;
2  TotalSlots = 8;
3  AllocatedSlots = 6;
4  FreeSlots = TotalSlots - AllocatedSlots;
5  Utilization = (AllocatedSlots/TotalSlots)*100;
6  disp('Frame Utilization (%)')
7  disp(Utilization)
8  figure
9  % Graph 1: Simple Line Plot (SAFE - no bar function)
10 subplot(2,1,1)
11 plot([1 2],[AllocatedSlots FreeSlots],'-o','LineWidth',2)
12 title('GSM Frame Utilization (Allocated vs Free)')
13 xlabel('Slot Type Index (1=Allocated, 2=Free)')
14 ylabel('Number of Slots')
15 grid on
16
17 % Graph 2: Pie Chart (SAFE)
18 subplot(2,1,2)
19 pie([AllocatedSlots FreeSlots])
20 title('Slot Distribution in GSM Frame')
21 legend('Allocated','Free')

Command Window

Frame Utilization (%)
    75

fx >>
```



Exp2.3

```

Editor - C:\Users\abhiz\Desktop\5g-6g-communication\exp2_obj_3.m
exp1.m  exp1_obj_2.m  exp1_obj_3.m  exp2_obj_1.m  exp2_obj_2.m  exp2_obj_3.m  +
1      clc; clear; close all;
2      Users = 1:8; % Total users
3      TotalSlots = 5; % Available slots
4      Alloc = zeros(1,8); % Initialize allocation
5      Alloc(1:TotalSlots) = 1; % Assign slots to first users
6      Blocked = 1 - Alloc; % Remaining users blocked
7      disp('User Wise Slot Allocation')
8      disp(table(Users',Alloc','VariableNames',{'User','Status(1=Allocated,0=Blocked)'}))
9      figure
10     % Graph 1: Allocation per user
11     subplot(2,1,1)
12
13     stem(Users,Alloc,'filled','LineWidth',2)
14     title('TDMA Slot Allocation per User')
15     xlabel('User Index')
16     ylabel('Slot Status (1=Allocated, 0=Blocked)')
17     ylim([-0.2 1.2])
18     grid on
19     % Graph 2: System view (safe alternative to bar)
20     subplot(2,1,2)
21     plot(Users,Alloc,'-o','LineWidth',2)
22     hold on
23     plot(Users,Blocked,'-s','LineWidth',2)
24     title('TDMA Slot Utilization Analysis')
25     xlabel('User Index')
26     ylabel('Status')

Command Window
Frame Utilization (%)
75
fx >>

```

